

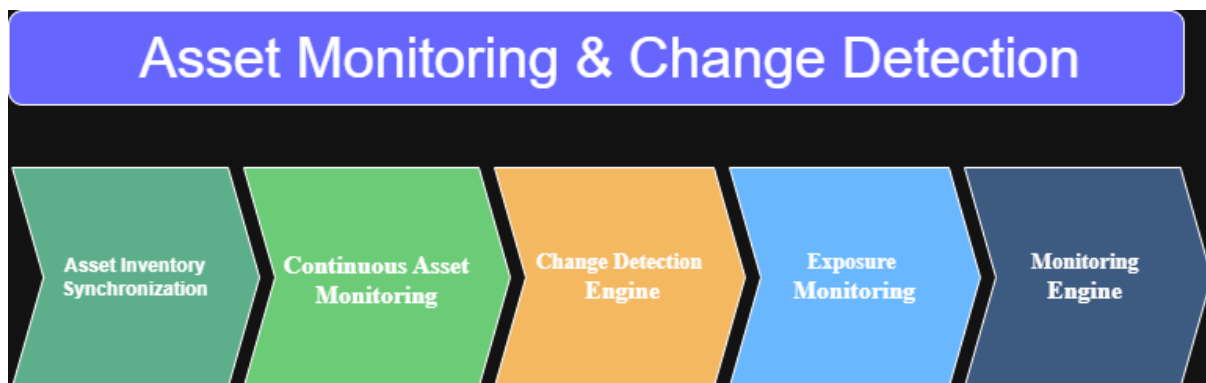


TEAM 2 – Asset Monitoring & Change Detection

Objective

Team is responsible for continuously monitoring discovered assets from the ASM module and detecting any infrastructure, configuration, exposure, or technology changes in real time. The goal is to create a Monitoring Engine that identifies newly exposed services, unauthorized modifications, shadow assets, risky changes, and suspicious behaviours across the attack surface.

Implementation Workflow



1. Phase 1 – Asset Inventory Synchronization

Purpose

The purpose of **Asset Inventory Synchronization** is to create and maintain a centralized, continuously updated inventory of all organizational assets discovered through the ASM system. It ensures that every monitoring, risk analysis, and attack path activity is based on accurate and current asset information.

Implementation Steps

- Receive discovered assets from ASM engine
 - Domains
 - Subdomains
 - IP addresses
 - Open ports
 - Cloud assets
 - Web technologies



- Store assets in centralized inventory database
 - PostgreSQL / TimescaleDB
 - Asset tagging
 - Business unit mapping
 - Environment tagging (Prod/UAT/Dev)
- Create baseline snapshot
 - DNS records
 - SSL certificates
 - Open ports
 - Running services
 - Web technologies
 - Cloud metadata

2. Phase 2 – Continuous Asset Monitoring

Purpose

The purpose of **Continuous Asset Monitoring** is to continuously track organizational assets and detect any changes, exposures, or suspicious activities across the attack surface in real time. This phase ensures that newly exposed services, infrastructure modifications, technology changes, and unauthorized assets are quickly identified before they become security risks.

Implementation Steps

- Schedule periodic scans
 - Every 1 hour → critical assets
 - Daily → full infrastructure scan
 - Weekly → deep enumeration
- Detect:
 - New subdomains
 - New exposed ports
 - New technologies
 - DNS changes
 - SSL expiry
 - CDN changes
 - Web application changes



- Compare results with previous baseline
- Trigger change events if deviations detected
- Tools
 - Wazuh - Centralized security monitoring and alerting
 - Osquery - Endpoint and system monitoring
 - Zeek - Network traffic analysis and monitoring
 - Suricata - Real-time threat and intrusion detection
 - Prometheus - Infrastructure and service monitoring
 - Grafana - Visualization and dashboarding

3. Phase 3 – Change Detection Engine

Purpose

The purpose of the **Change Detection Engine** is to identify, analyze, and classify any modifications occurring across the organization's attack surface by comparing current asset states with previously established baselines. This phase helps detect unauthorized, suspicious, or risky infrastructure changes before they can be exploited by attackers.

Implementation Steps

- Build baseline fingerprints
 - Port state baseline
 - HTTP response baseline
 - Certificate fingerprint
 - Technology stack fingerprint
- Run differential analysis
 - Compare latest scan with previous scan
 - Detect additions/removals/modifications
- Categorize changes
 - Informational
 - Suspicious
 - Critical exposure
- Assign severity score



Change Detection Example,

Change Type	Detection	Severity
New Port Opened	SSH opened on public IP	High
New Subdomain	staging.example.com	Medium
New Admin Panel	/admin exposed	Critical

4. Phase 4 – Exposure Monitoring

Purpose

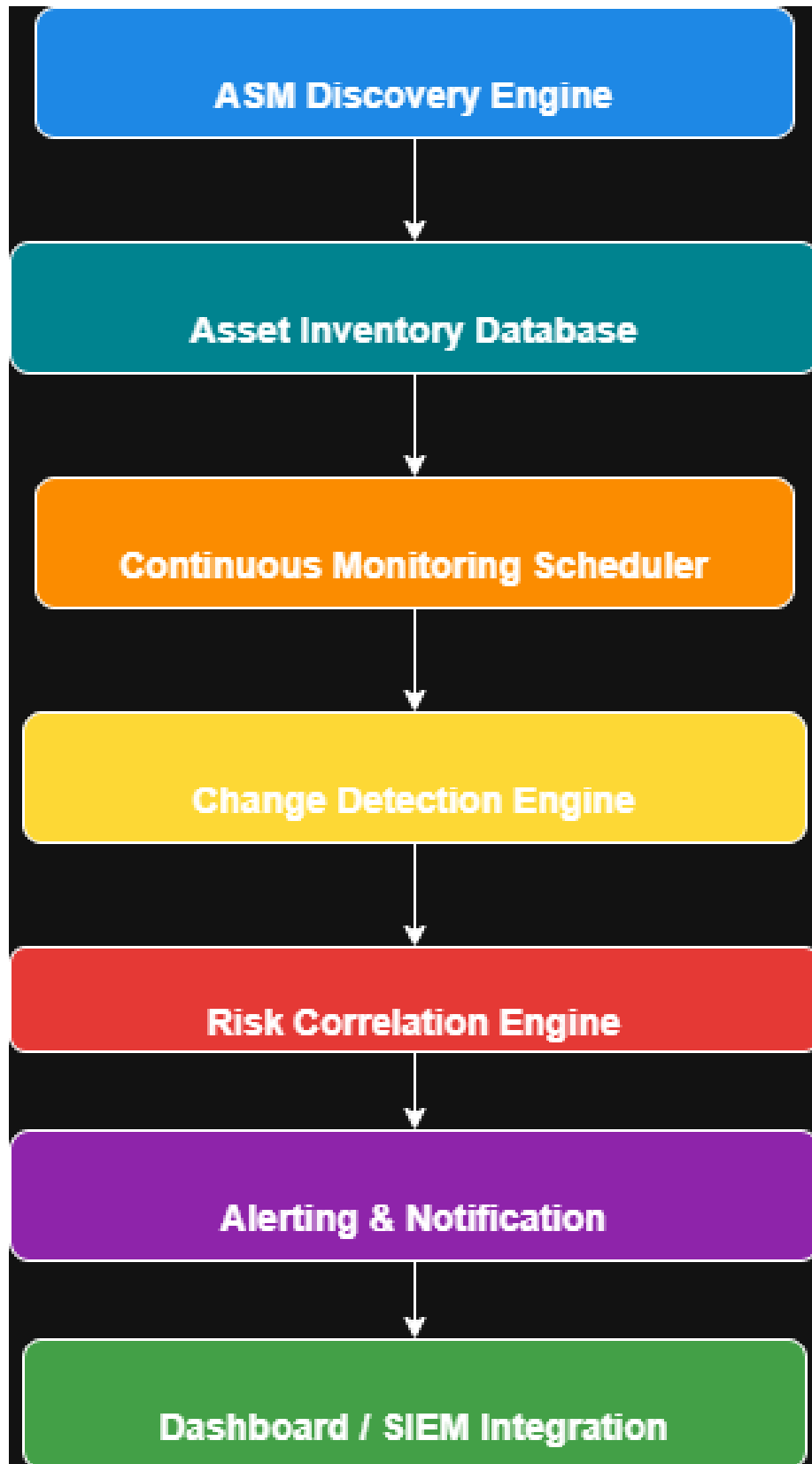
The purpose of **Exposure Monitoring** is to continuously identify and track publicly exposed assets, services, misconfigurations, leaked information, and internet-facing risks that could increase the organization's attack surface. This phase focuses on detecting security exposures before attackers can discover and exploit them.

Implementation Steps

- Monitor exposed services
- Detect public cloud exposure
- Monitor leaked credentials
- Monitor expired certificates
- Detect exposed databases
- Track shadow IT assets
- Tools
 - SpiderFoot - Automated attack surface and exposure discovery
 - theHarvester - External reconnaissance and information gathering
 - Censys - Internet exposure visibility
 - OpenCTI - Threat intelligence management and correlation
 - MISP - Threat intelligence sharing and IOC management

5. Monitoring Engine Architecture

The **Monitoring Engine Architecture** is designed to provide continuous visibility, real-time monitoring, change detection, exposure analysis, and automated alerting across the organization's attack surface. It integrates asset discovery, monitoring agents, network sensors, threat intelligence, risk analysis, and SIEM dashboards into a centralized CTEM monitoring platform.





Expected Deliverables

Monitoring Engine

Capabilities:

- Continuous monitoring
- Asset change tracking
- Configuration drift detection
- Exposure monitoring
- Real-time alerts
- Risk scoring support

Output Data Shared to Team 3

- Newly exposed assets
- Configuration changes
- Suspicious behaviour
- Exposure events
- Technology changes
- Threat intelligence correlation

Expected Outcome

The Asset Monitoring & Change Detection module is designed to provide continuous visibility into organizational assets and infrastructure changes across the attack surface. The primary outcome of this team is the development of a centralized monitoring engine capable of tracking asset modifications, detecting exposure changes, and maintaining real-time awareness of infrastructure status. This module acts as a bridge between asset discovery and risk analysis by ensuring that all identified assets remain continuously monitored and updated within the ThreatGraph platform.

The expected deliverable includes a scalable monitoring framework that supports continuous asset synchronization, historical change tracking, exposure monitoring, and automated alert generation. The module will also integrate with the central database and downstream risk-analysis components to ensure accurate and updated asset intelligence across the platform.